Multiple extensions of a quadruple

Problèmes ouverts — A. Dujella · 2026

Problem 1.2 (Dujella, 2026). Is there any positive integer $k \geqslant 2$ such that the Diophantine quadruple $\{k-1, k+1, 4k, 16k^3-4k\}$ has more than one extension to a rational Diophantine quintuple ?

Remark. Stoll (Acta Arith. **190** (2019), 239–261) proved that the extension is unique for $k = 2, 3, 4, 5$.